

CONCEPT SUMMARY: Properties of Logarithms

	Product Property	Quotient Property	Power Property	Change of Base
ALGEBRA	$\log_b(mn) = \log_b m + \log_b n$	$\log_b \left(\frac{m}{n}\right) = \log_b m - \log_b n$	$\log_b(m^n) = n \cdot \log_b m$	$\log_b m = \frac{\log_a m}{\log_a b}$
WORDS	The log of a product is the sum of the logs.	The log of a quotient is the difference of the logs.	The log of a number raised to a power is the power multiplied by the log of the number.	The log base b of a number is equal to the log base a of the number divided by the log base a of b .
NUMBERS	$\log_2(20) = \log_2(4) + \log_2(5)$	$\log_{10} \left(\frac{2}{3}\right) = \log_{10} 2 - \log_{10} 3$	$\log_3(16) = 4 \cdot \log_3 2$	$\log_5 7 = \frac{\log 7}{\log 5}$